

Mohamed Yousry

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SUMMARY

AI Engineer and Backend Developer with experience leading B2B and B2C platforms and shipping production AI systems, including agentic and multi-agent architectures (A2A, AG2), LLM deployments, and on-device AI. Strong background in Python, FastAPI, Docker, Redis, and PostgreSQL, with a focus on microservices, automation, and real-time data processing.

EXPERIENCE

Drive-Nova, Lead Engineer (Hybrid • UAE)

Apr 2026 - Present

- Lead engineering across B2B and B2C platforms (including managed Magento storefronts), owning architecture, delivery, and ongoing platform operations.
- Architecting and managing upcoming AI initiatives, including Agent-to-Agent (A2A) architecture and multi-agent frameworks such as AG2.
- Built and operate the e-commerce platform and AI behind TyresOnline (UAE), while managing integrations and technical relationships across partner platforms in the tyre industry, spanning Europe, Egypt, and the UAE.

Pillar Construction Group, Fractional CTO & Lead AI/Backend Engineer

May 2025 - Present

- Designed and delivered a full-stack AI procurement platform aggregating 50,000+ federal (SAM.gov) and SLED (HigherGov) opportunities, with multi-schema PostgreSQL, semantic search via embeddings, and an event-driven processing system.
- Built ELI, a conversational AI assistant with 24 specialized procurement tools (opportunity search, document Q&A, board management) powered by GPT-5/GPT-4o, plus a 4-tier matching pipeline (geographic, metadata, documents, LLM) and a PaddleOCR pipeline for scanned government forms.
- Architected microservices with FastAPI, Celery (6 specialized workers), Redis, and Supabase; deployed full production infrastructure with Docker, Coolify, Nginx, Cloudflare, GitHub Actions, and Sentry monitoring.

MyAly.AI, AI Engineer & Automation Developer

Feb 2025 - Apr 2025

- Built and maintained Stripe payment systems, including subscriptions, webhooks, and secure checkout, enhancing payment processing efficiency.
- Developed and integrated frontend flows with Backend APIs, improving website performance and user experience.
- Set up and managed infrastructure and deployments using domains, DNS, Nginx, and Git workflows, ensuring robust and fault-tolerant applications.

TotallyNot, Co-founder & AI Engineer

Sep 2024 - Mar 2025

- Led the development of AI-driven features for the Local on-device AI & Flutter, enhancing system performance and user experience.
- Designed and deployed scalable backend architectures using FastAPI, PostgreSQL, and Docker, improving data processing efficiency.
- Implemented CI/CD pipelines for seamless deployment, improving system stability.

Freelance, Freelance Developer

Jan 2020 - Present

- Architected and deployed large-scale production systems, including AI pipelines, OCR systems, mobile applications, web platforms, and automation bots, enhancing efficiency and scalability.
- Designed and implemented OCR and document-processing systems, handling image preprocessing, text extraction, validation, and structured data output, improving data processing.
- Managed full infrastructure lifecycle, including cloud setup, server provisioning, Dockerized deployments, CI/CD pipelines, domain/DNS configuration, and production monitoring, ensuring robust and fault-tolerant applications.

SELECTED PROJECTS

Roundtable — Self-Hosted Film & Series Tracking Platform

Open Source

- Built a self-hosted film and series tracker with behavior-based recommendations (per-user taste vectors with pgvector), per-country streaming availability, episode-level series tracking, cinema release alerts, and group watch nights with blended-taste voting.
- Added an AI assistant that does grounded web research over a film database before answering, and an MCP server that exposes watch history to external AI assistants.
- Stack: FastAPI, React, PostgreSQL, pgvector, TMDB API, Docker.

ileterate — AI-Powered Multilingual Grammar Checker

Open Source

- Built a privacy-first, self-hostable grammar and rewriting platform supporting Dutch, English, German, French, and Spanish, combining LanguageTool's rule-based detection with LLM corrections and a unique validation loop that re-checks LLM output to prevent AI-introduced errors.
- Cross-platform Flutter app (iOS, Android, Web, Desktop) with FastAPI backend, RSA + AES end-to-end encryption, API key authentication, and one-command Docker Compose deployment.
- Stack: Python, FastAPI, Flutter, LanguageTool, Docker, OpenAI-compatible LLMs.

CSphere — Multi-Agent AI IDE

In Development

- Architected and authored full system documentation for a VSCode-based IDE with an 11-agent system (orchestrator-driven, MCP-based) covering autonomous coding, debugging, testing, refactoring, and deployment, plus a free community marketplace for tools, agents, and extensions.
- Designed local-first infrastructure: SQLite plus ONNX Runtime embeddings running directly in Electron with no middleware, optional Supabase sync for auth and usage, and hosted inference for Llama 4, Qwen, and GPT OSS models.
- Stack: Electron, TypeScript, Python, FastAPI, SQLite, ONNX Runtime, Supabase, MCP.

strix-halo-unsloth — ML Infrastructure for AMD Strix Halo

Open Source

- Authored the working setup guide and prebuilt Podman containers for running PyTorch and Unsloth fine-tuning on AMD Ryzen AI Max+ 395 (gfx1151) with 128 GB unified VRAM; identified the kernel regression and ROCm version constraints that block other setups.
- Fine-tuned proof-of-concept models to validate the stack end-to-end.
- Stack: PyTorch (ROCm 7.1), Unsloth, LoRA, Podman/Docker, Linux kernel tuning.

EDUCATION

New Mansoura University

Bachelor's • AI Engineering • 2021 - 2026

GPA 3.3/4

SKILLS

Languages: Python • TypeScript • JavaScript • C++ • C • C#

Backend & APIs: FastAPI • Flask • Django • REST APIs • Microservices • Celery • Stripe

AI & ML: Large Language Models (LLMs) • Multi-Agent Frameworks (AG2, MCP) • LangChain • Llama.cpp • PyTorch • TensorFlow • Fine-tuning (LoRA, Unsloth) • PaddleOCR • NLP • Embeddings (pgvector) • ONNX Runtime

Mobile & Frontend: Flutter • React • Next.js • Electron

Databases: PostgreSQL • Supabase • Redis • MySQL • SQLite

DevOps & Automation: Docker • Nginx • Linux • Cloudflare • CI/CD (GitHub Actions) • Git • Sentry • n8n • ETL Pipelines